Smart Campus: linear energy forecasting

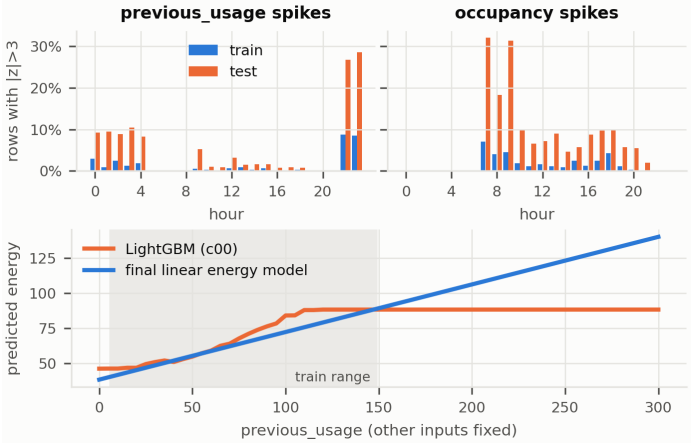
Cookie Squad | Final package c18 (c17 parameters) | CSV identical to c17: public RMSE **3.18775**
Best observed public: c16 3.18618 | Complete-row energy-model CV: 2.998

1 · Problem

Predict *energy_usage* for 12 buildings (8 types) from building, hour / weekday / month, temperature, humidity, occupancy and *previous_usage*; metric RMSE. Test rows arrive shuffled without ids, so features are row-local. Its previous-period usage has a daily profile matching energy at the **same** hour (residential corr. 0.97 vs 0.52 one hour back), consistent with a same-slot reference. No recoverable sequence was found; the generator is unknown.

2 · Critical insight: test is shifted, and the extremes are real

Test has 3× train's missing rate (18.2% vs 5.8% of rows) and, by a target-free consistency model (each feature predicted from the others), **16.6%** spike rows (|z|>3) vs **4.1%** in train, on a schedule (prev 22:00–04:00, occupancy 07:00–09:00). They are not errors: on train spike rows the energy left unexplained without the feature tracks its deviation (r = 0.92) along the ordinary-row line. So the energy equation must **extrapolate**; trees saturate.



3 · Validation

Energy model: paired 5-fold × 4-repeat CV (*tree baseline: 2 repeats*) scored **clean**, **test-like** (validation rows get masks sampled from the test file's joint patterns) and **shift-weighted** (spike rows ×4.05 to test frequency; a train-vs-test classifier, AUC 0.70, independently gave them ~3×). Imputers: a **label-free proxy** hides observed values in held-out public-test rows, re-imputes them and weights feature errors by energy slopes. Its natural-missingness slice checks harder rows. Neither proxy observes truly missing values or equals final energy RMSE.

4 · Method

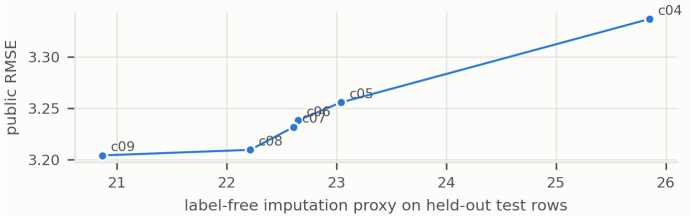
Energy model (ridge, α=3): per-building intercept, occupancy and temperature slopes; shared *previous_usage* (~0.34) and humidity slopes; smooth daily profile per type, residential day/night steps, shrunk hour corrections; weekend; season; temperature². **Missing inputs:** c03's per-pattern sub-models distort correlated slopes (without occupancy the prev slope inflates 0.34→0.46); **c04** imputes inputs and keeps the **full** model. **c05–c11** make the imputers test-aware: a train/test domain flag (predict as test), an average of 3 settings, test rows ×2, 5 copies of test-like mask augmentation, and **deviation features** (each observed input minus its building×hour and building mean), plus cyclic time and corrected mask-column ordering. LightGBM fills inputs; observed occupancy and previous usage keep their linear response.

c17 missing-value bias: naturally missing train rows have biased OOF residuals. Estimate a shift per feature in conditional-SD units, shrink toward zero by empirical Bayes, then add *kappa* × *slope* × *conditional SD* for missing inputs. Previous usage has the strongest support (raw kappa 0.418, SE 0.153; shrunk 0.297). No public score enters the correction rule.

5 · Results

RMSE	Clean CV	Test-like	Shift-wtd	Public
c00 LightGBM	3.341	3.964	4.281	–
c03 linear	3.077	3.296	3.543	3.36875
c04 + imputation	3.048	3.228	3.363	3.33681
c09 dev features				3.20392
c11 fix + schedule				3.19406
c16 best public				3.18618
c17 final statistics				**3.18775**

c04 beats c03 in 4/4 CV repeats on every view (test-like Δ -0.068, shift-weighted -0.180; MSE 95% CI [-0.67, -0.25]). Test-aware imputation then cut public RMSE 3.33681 → 3.20392. The proxy ranked c04–c09 correctly (below), but its gain sizes transferred inconsistently. Final c18 uses c17's train-estimated rule; c16's offsets were partly selected on public scores. Their 0.00157 gap is too small to establish a private-test difference.



6 · Error analysis: what is left

Rows without missing values (82% of test) are at the noise floor (~3.0): LightGBM, ExtraTrees, CatBoost and an MLP trained on out-of-fold residuals reach R^2 below 0, and group slopes, spike-specific slopes, density weighting and blends did not help. Forensics found no sequence, duplicates or cross-row linkage. The remaining error sits in rows with a missing value, where hidden spikes are hard to reconstruct. New five-fold extrapolating imputers worsened the proxy (20.81→21.26). An energy-weighted imputer blend improved it slightly (20.81→20.73), but its paired 95% MSE interval [-0.177,+0.027] includes zero; neither replaced the final model.

7 · Campus insights (associations, not causal effects)

+100 occupants ~ **+13.4** energy units in the lecture hall vs **+4.5** in residences. Science buildings carry the largest modelled base load (39–43 vs 16–27), consistent with fixed equipment loads (interpretation). Residences step down at 08:00 and up at 21:00. Uses: hourly peak planning and flagging buildings that exceed expected use.

8 · Private execution and limitations

c18 recovers types from known building IDs, isolates failed imputers and has linear/mean fallbacks. Its CSV is byte-identical to c17. Fresh kernels and a clean pinned-library environment passed; all 16 missingness patterns across 12 buildings were checked. Complete-row RMSE (~3.0) supports limited headroom, not proof of an irreducible floor. Private-domain and MNAR-mechanism transfer remain assumptions; MNAR cross-fitted MSE gain is uncertain (-0.078, SE 0.126). Temperature's quadratic term makes mean imputation approximate. Source: raw files, frozen packages and experiments/NOTES.md (rounds 1–15).